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SAARTHI'26 Submission

Smart Cities & Public Safety

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College Name: Graphic Era Hill University ,Dehradun ,Uttarakhand

Team Name

Problem Statement

Problem Statement:

- Cities run traffic, emergency, infrastructure and civic services as separate systems. When a flood, crash or congestion event hits, each department sees only its own alert, with no shared, verified picture of what is happening and no coordinated response. Ambulances are routed on conditions as they are now, not as they will be on arrival, and citizens struggle to get reliable updates or track their complaints.

Detailed description about the problem:

- **Disconnected departments:** traffic, emergency, infrastructure and civic teams each hold their own data and raise isolated alerts.
- **Reactive traffic management:** routing and signals react to current congestion instead of forecasting conditions when the vehicle arrives.
- **Unreliable incident context:** reports lack a common format, timestamp and confidence, so conflicting information is hard to reconcile.
- **Fragmented citizen communication:** complaints, notices and updates are spread across channels with no single tracking ID.



Proposed Solution

Solution Statement:

City Sense is an agentic operating system for smart cities. Every observation is normalised into one shared city context. Specialist agents (Traffic, Emergency, Infrastructure/Disaster, Environment, Transit, Civic) read it and publish findings. A city coordinator links them into one incident, ranks urgency and proposes an action plan. Low-risk notifications can run automatically, while high-impact actions such as signal control, road closure, dispatch or evacuation wait for explicit operator approval.

Detailed Solution description:

- **Shared city context:** roads, hospitals, incidents, sensors and complaints on one 2D and 3D map.
- **Specialist agents:** each publishes a structured finding or proposed action with evidence and confidence.
- **City coordinator:** links related events into one incident with a unified timeline.
- **Predictive emergency routing:** arrival-time-aware ETAs, committed routes and evidence-driven switching.
- **Civic avatar and complaints:** answers only from a verified knowledge base; geo-tagged complaint tracking.
- **Human oversight:** every approval or rejection is audit-logged.

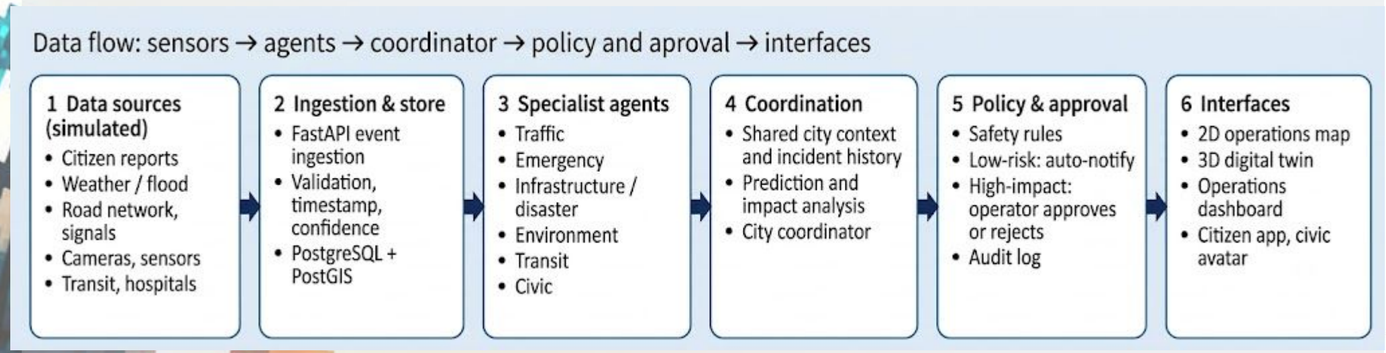
Citywide feature set:

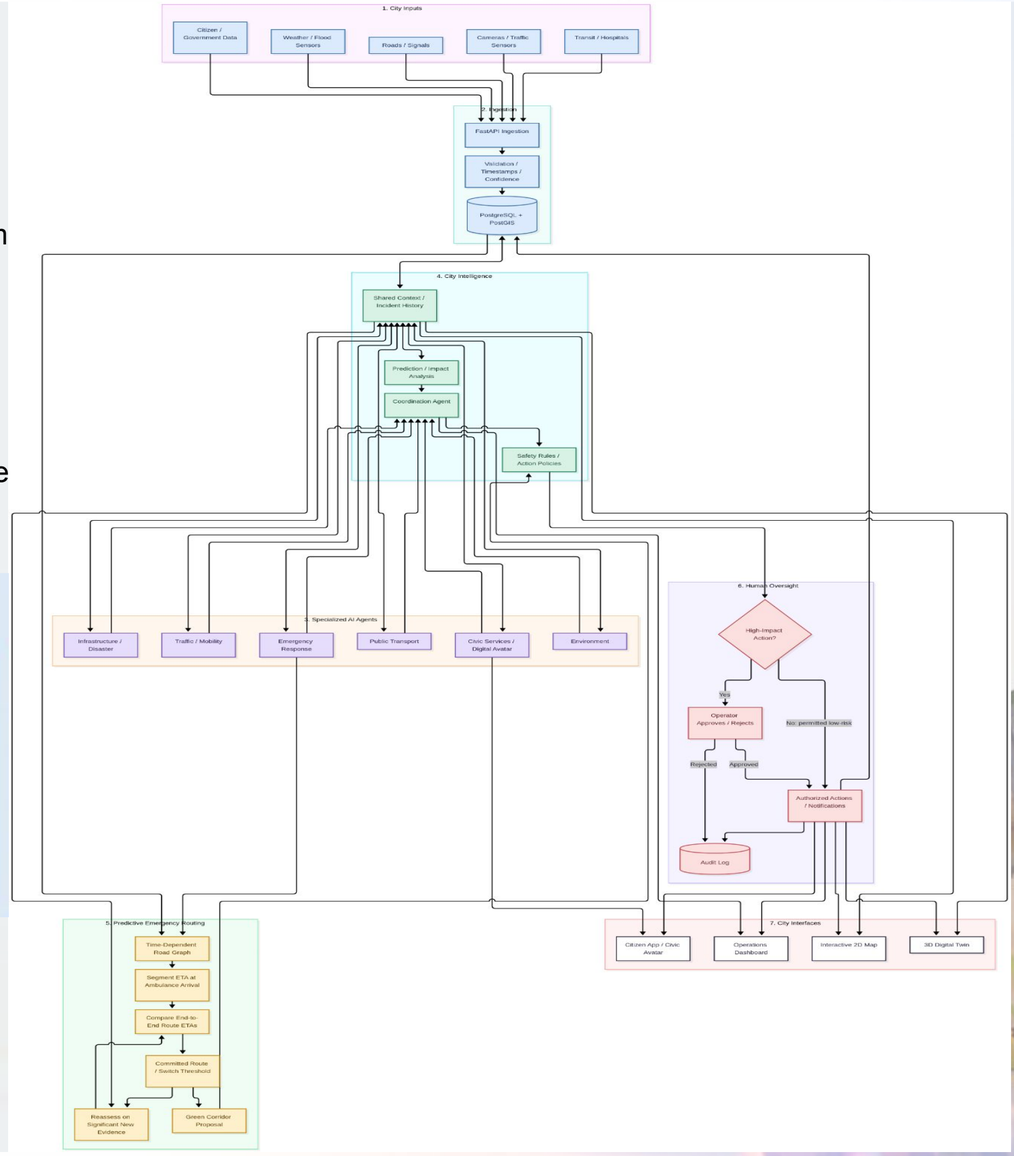
- **In prototype (simulated data):** Emergency & disaster, Traffic & mobility, Infrastructure, Public transport, Environment, Civic avatar, Complaint tracking.
- **Planned (not built):** Governance updates, Townhalls.

Technical Approach

Technologies To Be Used:

- **Frontend:** React, TypeScript, Vite
- **Maps and visualisation:** MapLibre GL for the 2D map, Three.js for the 3D city view
- **Backend:** Python, FastAPI, REST for queries and actions, WebSockets for live map updates
- **Data:** PostgreSQL with PostGIS for roads, zones, incidents, assets and complaints, using seeded dem data
- **Routing and simulation:** NetworkX road graph, with SUMO only if time permits
- **Agents:** Traffic, Emergency, Infrastructure/Disaster, Environment, Transit and Civic agents, plus a city coordinator, built as deterministic rule-based services
- **Civic AI:** LLM with RAG over a small verified local-government FAQ, with a deterministic fallback
- **Perception:** sample camera and sensor events (OpenCV/YOLO only if a real video source is available
- **Deployment:** Docker Compose for the frontend, API and database

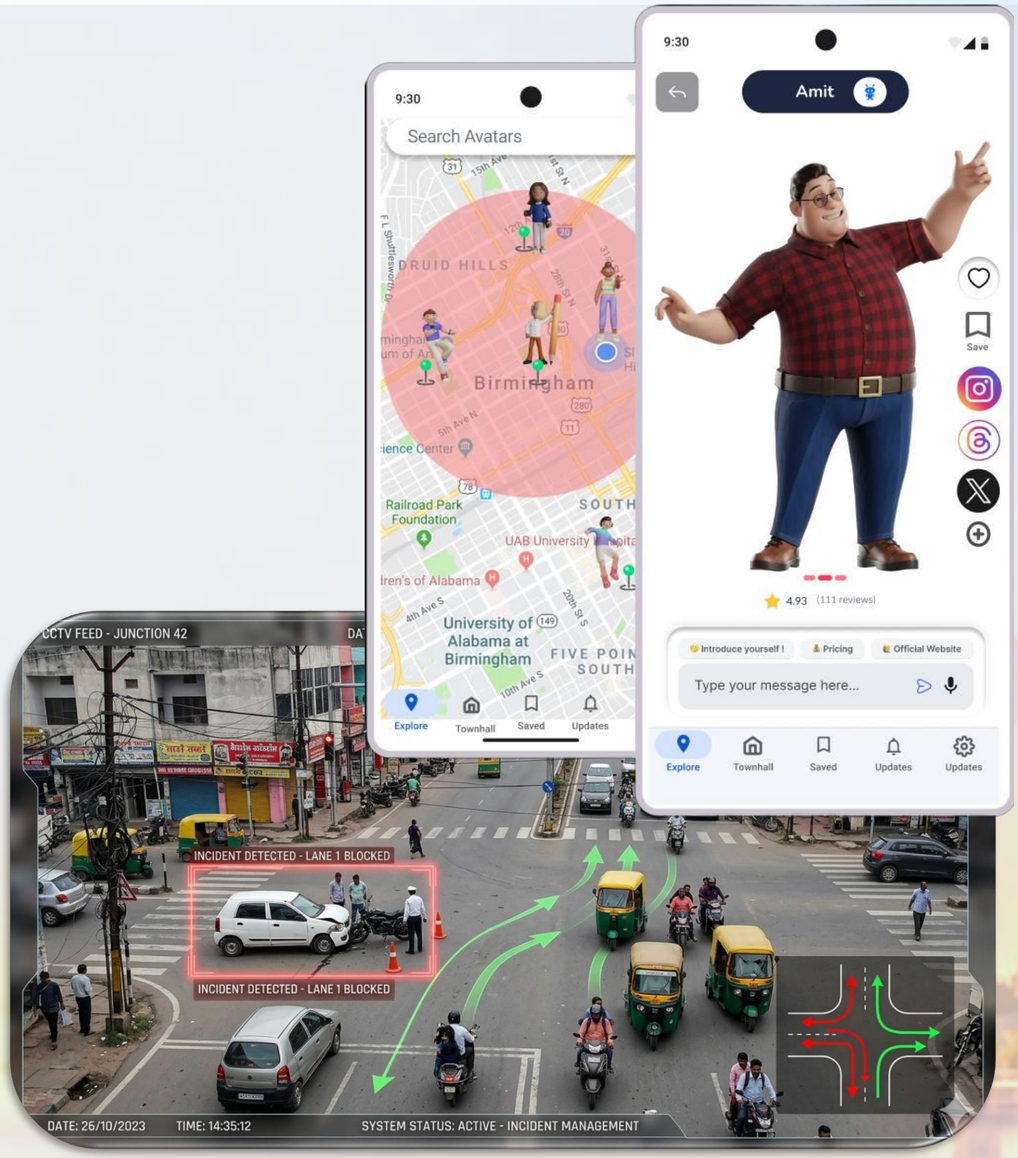




Detailed Description:

Every city domain on one shared map, coordinated by cooperating AI agents

Domain	What it does
Emergency & disaster	Predictive ambulance routing with committed routes and evidence-driven switching; flood and crash response; green-corridor proposals
Traffic & mobility	Congestion forecasting, incident impact analysis, signal-timing proposals
Public transport	Bus and transit disruption alerts, route-impact analysis
Infrastructure	Drainage, road and underpass risk monitoring; work-order proposals
Environment	Weather, rain and flood readings that feed the other agents
Civic avatar	Multilingual assistant that answers only from a verified knowledge base and cites its source
Complaint tracking	Geo-tagged complaints with a tracking ID, routed to the responsible department with live status
Governance & escalation	Complaint → worker assigned → warnings if unresolved → after 5 warnings, escalates to higher authority → continues level by level up to the CM portal



Impact and Benefits

Impact Metric	Details
Social Impact	Improves access to government schemes through multilingual AI avatars. Enables citizens to report civic issues, receive verified information, and engage with local authorities. Improves emergency response and public safety.
Economic Impact	Reduces fuel wastage and time lost in traffic congestion. Improves transport efficiency, supports local businesses, and helps emergency services reach destinations faster. Better visibility of businesses and government initiatives.
Environmental Impact	Reduces vehicle idling, fuel consumption, and associated emissions through adaptive traffic signals and intelligent route optimization.
Target Audience	Urban citizens, commuters, ambulance and fire services, municipal authorities, traffic police, local businesses, and city administrators.

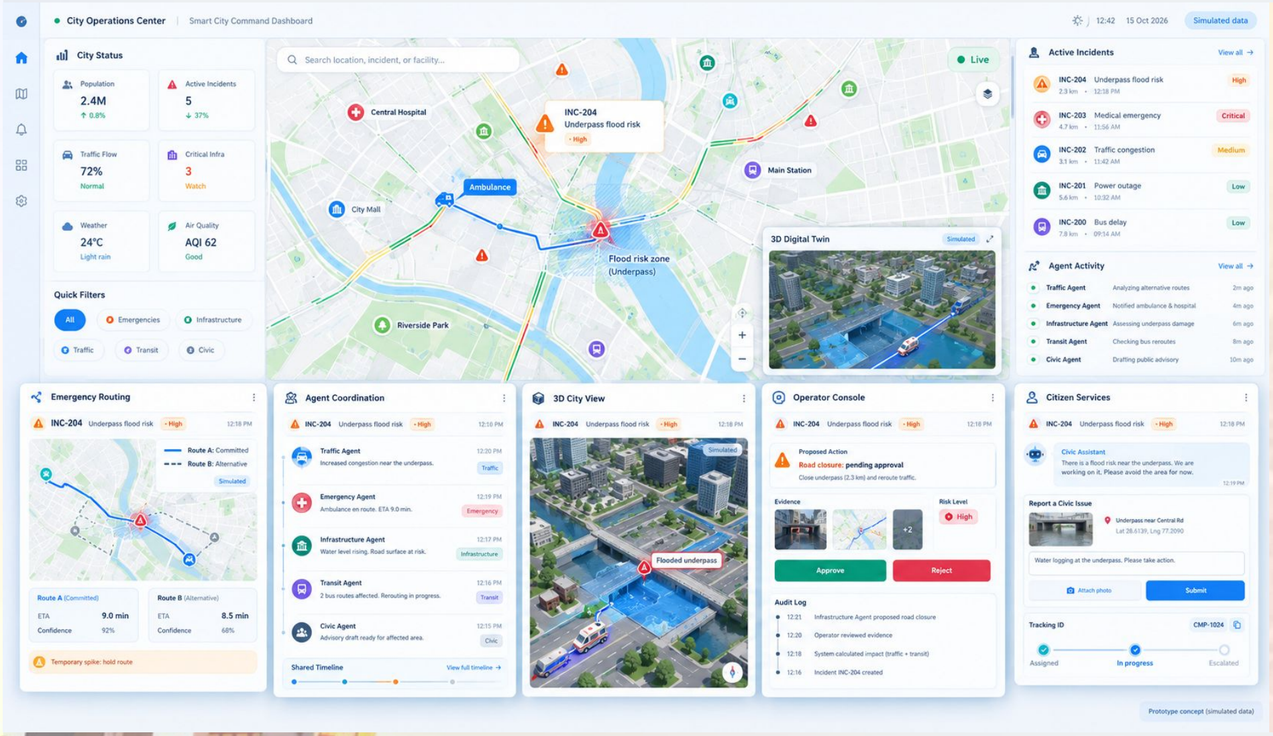


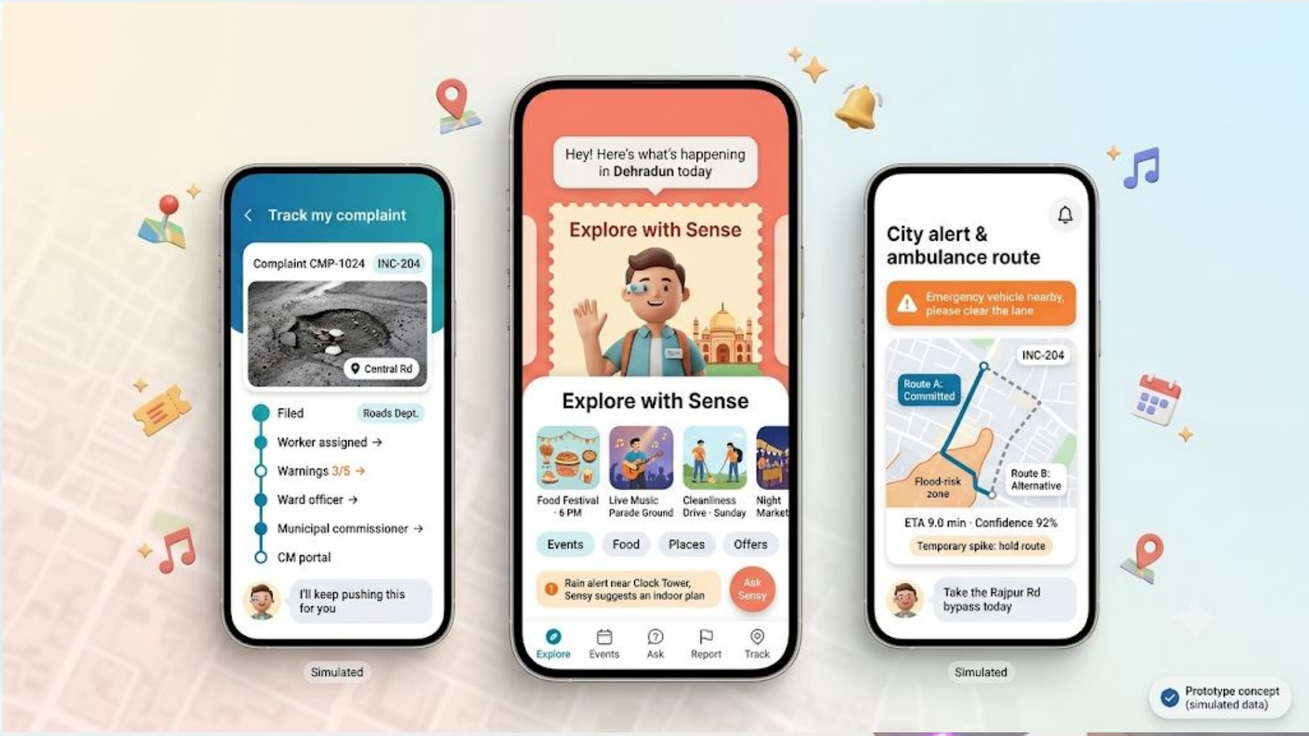


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Research & References

Research Work:

Research basis (how each source is used):

- **[1] Wei et al., *PressLight: Learning Max Pressure Control to Coordinate Traffic Signals in Arterial Network*, KDD 2019.** Basis for our RL signal-control design and baseline.
- **[2] *Traffic Signal Control With Deep Reinforcement Learning* (IEEE 8054579).** DQN-based signal control.
- **[3] *Deep Reinforcement Learning for Traffic Light Control in Vehicular Networks* (IEEE 9098073).** RL in connected-vehicle settings.
- **[4] Lopez et al., *Microscopic Traffic Simulation using SUMO*, ITSC 2018.** Simulation environment for training and benchmarking.
- **[5] Lewis et al., *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*, NeurIPS 2020.** Foundation for grounded, citable avatar answers.
- **[6] RAG avatar memory paper (ScienceDirect, S2666188825006586).** Avatar memory design.
- **[7] Constitution of India, Eighth Schedule.** Scope of language support (22 scheduled languages).

Links:

- Sarvam AI docs: https://docs.sarvam.ai/sarvam-api-docs
- SUMO: https://www.eclipse.org/sumo/
- Ultralytics YOLO: https://github.com/ultralytics/ultralytics
- RAG paper: https://arxiv.org/abs/2005.11401
- PressLight: https://dl.acm.org/doi/10.1145/3292500.3330949
- IEEE 8054579: https://ieeexplore.ieee.org/document/8054579
- IEEE 9098073: https://ieeexplore.ieee.org/document/9098073
- OpenStreetMap: use the OSM wiki or Overpass API page you actually call, not taginfo.